

Antisaccadic Training to Improve Impulsivity in Binge Eating Disorder

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Abstract

Patients with binge eating disorder (BED) show generally increased impulsivity and especially increased food-related impulsivity. Both are closely linked to the core pathology of BED, which relates to regular binge eating episodes with experienced loss of control. The antisaccade task is an established paradigm assessing response inhibition as a pivotal component of impulsivity. It requires participants to execute antisaccades; that is, they are supposed to look in the opposite direction of a stimulus that automatically catches attention by appearing in the peripheral visual field. High rates of prosaccades to the peripheral stimuli are considered indicators of increased impulsivity. Presenting food pictures as peripheral stimuli, this task can be used to investigate food-related impulsivity. We propose modifications of this task in order to design it as an antisaccadic training in which BED patients practise the suppression of food-related responses, which should result in enhanced control over their eating behaviour. Copyright © 2013 John Wiley & Sons, Ltd and Eating Disorders Association.

Keywords

antisaccade; binge eating; eating disorder; impulsivity; training

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Introduction

Binge eating disorder (BED) is now listed as a unique psychiatric condition in the new edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5; American Psychiatric Association, 2013). According to DSM-5, recurrent episodes of overeating represent the core symptom of BED (American Psychiatric Association, 2013). During these so-called binge eating episodes, affected patients experience a loss of control over their eating behaviour and consume large amounts of food in a short period, often resulting in overweight and obesity. Impulsivity is currently discussed as a variable potentially contributing to this particular eating behaviour in BED, as this personality trait is closely linked to both reward processing and behavioural control (Dawe & Loxton, 2004; Schag, Schönleber, Teufel, Zipfel, & Giel, 2013).

Major treatment aims in BED are to reduce binge eating episodes and to regulate the eating behaviour. According to current treatment guidelines, psychotherapy—and specifically cognitive behaviour therapy—is recommended to achieve these treatment aims (Herpertz, Herpertz-Dahlmann, Fichter, Tuschen-Caffier, & Zeeck, 2011; National Institute for Clinical Excellence, 2004). To complement these current psychotherapy approaches in BED, interventions specifically targeting impulsivity might be crucial. Such an *add-on* intervention could be an antisaccadic training.

In the following sections, we will give a brief overview on the role of impulsivity in BED, we will introduce the antisaccade task

and we will outline the relation between antisaccade performance and impulsivity. Finally, we will sketch how an antisaccadic training in BED could look like and how it could contribute to current treatment approaches in eating disorders.

Impulsivity in binge eating disorder

Impulsivity is considered to be a multidimensional trait variable (e.g. Eysenck, 1967; Gray, 1970; Zuckerman, 1994) that is composed of two main components (Dawe & Loxton, 2004): (i) *reward sensitivity*, that is, the heightened drive towards pleasurable stimuli; and (ii) *disinhibited* and *rash-spontaneous behaviour* without considering possible consequences.

There is evidence for generally increased impulsivity in BED patients. Clinically, this is suggested by the high comorbidity of BED with other disorders that are associated with high impulsivity, such as substance use and attention deficit/hyperactivity disorder (Schmidt, Korber, de Zwaan, & Mueller, 2012). Moreover, BED patients report increased trait impulsivity in questionnaires (e.g. Davis et al., 2008; Schienle, Schäfer, Hermann, & Vaitl, 2009). In line with this, there is first experimental evidence that BED patients show increased sensitivity to various rewarding stimuli (Manwaring, Green, Myerson, Strube, & Wilfley, 2011) and increased disinhibited behaviour towards neutral stimuli (Balodis et al., 2013; Nederkoorn, Braet, Van Eijs, Tanghe, & Jansen, 2006).

However, specifically food-related impulsivity could considerably account for the eating behaviour seen in BED. The interplay

of both impulsivity components might result in binge eating episodes with loss of control, as BED patients might be hyper-responsive to food, which elicits an excessive reward response, and, as a consequence, they experience food cravings, start eating and finally lose control over food intake (Dawe & Loxton, 2004). Results from both observational studies investigating food intake and experimental studies investigating food cue processing confirm this assumption (see Schag et al., 2013, for a recent review): There is considerable evidence that BED patients show increased reward sensitivity towards food stimuli, and, although far fewer studies have been conducted on the second impulsivity component, there is also initial evidence for increased disinhibited behaviour towards food stimuli in BED patients. Notably, this latter impulsivity component seems to discriminate between the groups of obese individuals with and without BED (Schag et al., 2013) and therefore suggests that BED might represent a specific phenotype of increased food-related impulsivity in obesity (Carnell, Gibson, Benson, Ochner, & Geliebter, 2012; Schag et al., 2013). Individuals of this phenotype could therefore benefit from tailored interventions targeting this specific impulsivity deficit. Amongst others, such an intervention to specifically improve disinhibited behaviour by increasing inhibitory control in BED could be based upon the so-called antisaccade task.

The antisaccade task

The antisaccade task is an established experimental paradigm originally developed for the investigation of neurodegenerative conditions in neurology, which has by now become increasingly popular in cognitive neuroscience and the study of psychiatric disorders (Antoniades et al., 2013; Hutton & Ettinger, 2006). Although there are several variations of the antisaccade task and there have been recent attempts for an international standardization (Antoniades et al., 2013), all antisaccade protocols have two components in common (Figure 1a): Each antisaccade trial starts with a fixation cross presented in the centre of the computer screen. It is succeeded by a stimulus appearing in a slightly peripheral position to the left or right of the central fixation cross.

Participants are instructed to turn their gaze as quick as possible in the opposite direction or the exact mirror position of the stimulus as soon as it appears on the computer screen.

In the classical antisaccade task, the peripheral stimulus consists of a simple geometrical shape, for example, a circle or triangle (Munoz & Everling, 2004). Recently, several studies investigating psychiatric conditions adapted the classical version of the task by using complex disorder-related stimuli, such as emotional faces in anxiety (Mueller, Hardin, Mogg, et al., 2012; Wieser, Pauli, & Muhlberger, 2009). Accordingly, our workgroup has developed an adapted antisaccade task using food and nonfood stimuli as a research tool in eating disorders and obesity (Figure 1b) and uses eye tracking to assess gaze behaviour in this task.

Antisaccade performance

As outlined earlier, the antisaccade task requires the participant to perform so-called antisaccades, that is, to deliberately look in the opposite direction of a simple stimulus that is newly presented in the peripheral visual field. The prepotent reaction to such a novel visual stimulus is a gaze shift towards this stimulus, that is, the performance of a so-called prosaccade (Munoz & Everling, 2004). However, in order to meet task requirements, the participant has to firstly suppress a reflexive prosaccade to the stimulus and to secondly initiate and execute the adequate motor response—the antisaccade (Munoz & Everling, 2004). The antisaccade task is hence considered as a paradigm that allows insight into the volitional control of action (Hutton & Ettinger, 2006; Munoz & Everling, 2004), or, as Antoniades et al. (2013) put it, it provides ‘a window onto the operation of the very highest levels of cerebral function’ (p. 2).

Outcome measures of the antisaccade task

On an operationalization level, antisaccade performance is determined by several outcome parameters: Error rate is recognized as main task result, which is mostly expressed as percentage of trials in which the participant has performed an erroneous prosaccade

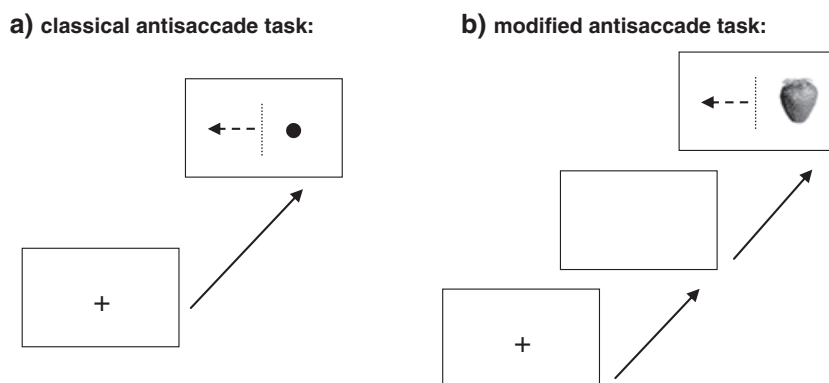


Figure 1. Schematic presentation of (a) the classical antisaccade task and (b) a modified food-related antisaccade task. Dashed arrows indicate the position in which the participants are required to turn their gaze to. A major variation of the antisaccade task lies in the sequence of the fixation cross and stimulus: (i) They can temporarily overlap, (ii) the fixation cross can be masked directly before stimulus onset or (iii) there can be a short gap showing a blank screen in between fixation cross and stimulus, which increases task difficulty. In the modified version, we have inserted such a gap. We have used food and nonfood pictures as peripheral stimuli and instructed participants to look at the opposite position as quickly as possible after stimulus onset

(Smyrnis, 2008). Average error rate in healthy participants lies around 20% (Hutton & Ettinger, 2006). Moreover, different latency measures can be derived, including the time required to perform an antisaccade or an erroneous prosaccade (Smyrnis, 2008). A further outcome category refers to measures associated with corrective behaviour, for example, the rate of corrective antisaccades after a preceding erroneous prosaccade.

Antisaccade performance as a measure of impulsivity

The antisaccade task requires participants to suppress an undesirable behaviour, which is highly dominant in the respective task situation. The task is therefore considered to assess a pivotal aspect of inhibitory control, the so-called *prepotent response inhibition* (Aichert et al., 2012). Translated on a performance level, participants who have impaired inhibitory control show more impulsive reactions by conducting erroneous prosaccades, which results in higher error rates. Beyond face validity, there is convincing evidence, for example, from neuropsychology and brain imaging, that antisaccade performance provides insights into impulsivity.

In a large sample of healthy participants, higher self-reported impulsivity was significantly associated with higher error rates, even after controlling for gender, age and intelligence (Aichert et al., 2012). In the same study, antisaccade performance was found to be significantly associated with performance in another established task assessing prepotent response inhibition, the so-called go/no-go task (Aichert et al., 2012). Brain imaging studies investigating neural correlates of saccadic control have identified a complex brain network activated during antisaccade performance. This network comprises amongst others brain areas typically involved in volitional behavioural and inhibitory control, such as the dorsolateral prefrontal cortex [see Hutton and Ettinger (2006) and McDowell, Dyckman, Austin, and Clementz (2008) for reviews]. Moreover, impaired antisaccade performance has been reported in different mental disorders that are known to be associated with overall increased impulsivity or specific problems in cognitive and behavioural inhibitory processes, including obsessive-compulsive disorder (Lennertz et al., 2012), attention deficit/hyperactivity disorder (Loe, Feldmann, Yasui, & Luna, 2009) and bipolar disorder (Mueller, Ng et al., 2012).

Antisaccade performance and food-related impulsivity in binge eating disorder

In a recently completed cross-sectional study, our workgroup has examined the role of both impulsivity components in BED. While we have used an established paradigm predominantly assessing reward sensitivity (Castellanos et al., 2009; Giel et al., 2011a), we newly developed an adapted version of the antisaccade task to investigate food-related impulsivity (Figure 1b). As compared with that in control groups of obese persons without BED and normal-weight individuals, antisaccade performance in BED patients was generally decreased and, in some measures, selectively impaired for food stimuli. This indicates increased rash-spontaneous behaviour in the BED group, which is partly even more pronounced when confronted with food. Additionally, antisaccade performance in the BED group was significantly correlated with self-reported trait impulsivity. The findings support

the view that BED represents a specific and distinct subgroup of obese persons with special deficits in impulsivity, which should be addressed in special interventions. Moreover, our workgroup has first longitudinal data from a pilot sample of BED patients suggesting that antisaccade performance tends to improve after cognitive behaviour therapy for BED and correlates with overall therapy outcome. Notably, reward sensitivity was hardly affected post-treatment in this patient sample. This initial evidence leads us to the cautious assumptions that disinhibited behaviour towards food is generally alterable in BED and—particularly being the more behaviour-related impulsivity component—might be the more promising starting point than the reward-related component to change impulsivity deficits in this patient group.

Antisaccadic training

An antisaccadic training is based on the antisaccade task and modifies the original experimental protocol in order to improve task performance. Similar training tools for mental disorders, which are based on experimental paradigms, have earlier been developed in other domains and have proven effective. The probably most widely studied approach is the so-called attentional bias modification (ABM; Beard, Sawyer, & Hofmann, 2012). The preferential allocation of attention to disorder-related information, which is referred to as an attentional bias, has been demonstrated in many mental disorders, including anxiety (Bar-Haim, Lamy, Pergamin, van Ijzendoorn, & Bakermans-Kranenburg, 2007) and eating disorders (Brooks, Prince, Stahl, Campbell, & Treasure, 2011; Giel et al., 2011b; Renwick et al., this volume). In ABM protocols, patients learn to redirect their attention away from disorder-related stimuli. A recent meta-analysis confirms the efficacy of ABM, especially for threat stimuli in anxiety (Beard et al., 2012). However, evidence for the modification of an attentional bias for appetitive stimuli—the category that would also include food stimuli—is inconclusive (Beard et al., 2012).

Although the application of ABM might also prove fruitful in the field of eating disorders, impulsivity deficits demonstrated in BED require a different category of training approach. This approach should emerge from a cognitive and/or behavioural control task, such as the antisaccade task. A recent experimental study using a different experimental task in healthy individuals basically supports the idea that eating behaviour can be influenced by practising response inhibition: Houben and Jansen (2011) have created a specific go/no-go task in which groups of chocolate cravers either had to consistently inhibit (no-go) or had to consistently exhibit (go) a manual response to chocolate pictures, whereas a third group received a mixture of both response types (control). As assessed by subsequent free chocolate consumption, the no-go training had resulted in increased inhibitory control, as participants of this task condition consumed significantly less chocolate compared with those of the control group.

The antisaccade task as a training tool

The antisaccade task has a number of advantages over other established response inhibition tasks [see Mueller, Hardin, Korelitz, et al. (2012) for a recent outline], some of which especially qualify it as a training tool. In contrast to most other paradigms in the domain of cognitive and behavioural control

(e.g. the go/no-go task), the antisaccade task does not require a manual response but is solely based on oculomotor performance. This makes it a more direct and ecologically valid measure of inhibitory control. Moreover, the neurobiological circuitry underlying antisaccade performance is well documented (Hutton & Ettinger, 2006), and there is firm evidence of specific impairments in a range of mental disorders, serving as a background for the interpretation of novel findings and the advancement of task-related studies. However, an essential prerequisite for the suitability as a training tool is the plasticity of antisaccade performance. Dyckmann and McDowell (2005) have demonstrated that healthy adults who received a regular antisaccade training significantly improved their error rates after one week. A further characteristic that makes the antisaccade task particularly suitable to be used as a training tool is the proven susceptibility for feedback and incentives (Mueller, Hardin, Korelitz, et al., 2012).

Jones and Field (2013) have recently developed an inhibition training based on an antisaccade protocol to influence drinking behaviour but found no effects of this training on alcohol consumption in a nonclinical sample of social drinkers. However, the authors have rather used a classical antisaccade task; that is, participants were not required to inhibit an oculomotor reaction to a critical alcohol-related cue, but to a simple target cross, while alcohol-related stimuli were presented as cues in a central position previous to the actual oculomotor task. To the best of our knowledge, no antisaccadic training has yet been developed to target symptoms in clinical samples and mental disorders.

Antisaccadic training in binge eating disorder

The basic idea of the antisaccade training is to influence disorder-related behaviour via the improvement of antisaccade performance, which has been demonstrated to be associated with the core pathology of BED. Specifically, BED patients practise the suppression of a food-related oculomotor response in this training, which should result in increased inhibitory control and should ultimately help them to exert a better control over their eating behaviour. On a task level, the aim of the training is to reduce the error rate of patients, that is, the number of prosaccades towards food stimuli.

For the BED-specific antisaccadic training, we propose as main modification of the food-related antisaccade task (Figure 1b) to exclusively use food stimuli. While BED has been associated with generally increased impulsivity as well as with specifically increased food-related impulsivity, it is well conceivable that affected patients might also benefit from a training targeting general inhibitory control, thus using stimuli that are unrelated to the disorder. However, disinhibition towards food is closely linked to the core pathology of BED, which suggests that a food-specific training might have larger effects and might more directly influence dysfunctional eating behaviour. This hypothesis is supported by a recent study in healthy participants where a general inhibitory training with a different neuropsychological task did not result in changes in eating behaviour (Guerrieri, Nederkoorn, & Jansen, 2012). In order to offer a training that is as relevant to the core pathology of BED as possible, the food stimuli used should predominantly consist of typical binge food, as these stimuli will elicit stronger cravings and disinhibition in affected patients. Whenever possible, the antisaccadic training could even be created as a partly adaptive tool, either by

integrating stimuli depicting the respective patient's individually preferred binge food(s) or by selecting food stimuli from the stimulus pool that the patient has previously rated highest, for example, for palatability or individually experienced cravings. As in the original version of the antisaccade task, the patients' task in the training will be to look at the opposite direction of the food stimulus as quickly as possible after stimulus onset. To increase task difficulty, we recommend inserting a gap in between the fixation cross and the food stimulus (Figure 1b). In order to allow patients an estimation of their individual progress, each training session should end with a feedback on their error rate.

Evaluating the appropriate dosage of the training required to attain meaningful effects is difficult because of the general lack of longitudinal data on antisaccade performance: Healthy individuals are able to improve their antisaccadic performance in a classical training with non-specific stimuli within up to four weekly sessions (Dyckman & McDowell, 2005). Most studies on inhibition trainings that are based on similar experimental protocols as the proposed antisaccadic training report results from a single session (Guerrieri et al., 2012; Houben & Jansen, 2011; Jones & Field, 2013); however, they were conducted in healthy or subclinical populations. We consider BED patients to need repeated sessions for task improvement, as the stimuli used are disorder-relevant and much more emotional and motivational meaningful as neutral stimuli in the classical antisaccade protocol and as they are a group of individuals with potentially additional problems in impulsivity, attention or working memory.

Perspectives

We have proposed several modifications of a food-related antisaccade task in order to design it as an antisaccadic training that offers BED patients the possibility to practise the suppression of food-related oculomotor responses. However, the feasibility of this training is yet to be evaluated in pilot studies, and its efficacy has to be investigated in controlled trials. If the antisaccadic training proves useful and effective, it is not expected to be a stand-alone therapy, but it might be a useful enhancing tool for current structured multimodal psychotherapy approaches that offer a comprehensive treatment of BED. As such an add-on, the training could especially complement those interventions targeting impulsive eating and the increase of inhibitory control, especially with regard to eating behaviour, and might ultimately contribute to the reduction of binge eating episodes. Beyond these potential symptom-related effects, the antisaccadic training might also enhance the overall effects of the comprehensive psychotherapy programme it is combined with, for example, by generally increasing flexibility. Moreover, beneficial effects of non-invasive brain stimulation techniques have been shown on tasks involving cognitive control in general (Wolkenstein & Plewnia, 2013) and antisaccade performance in particular (Crevits, Van den Abbeele, Audenaert, Goethals, & Dierick, 2005; Muri & Nyffeler, 2008), suggesting that the combination of antisaccade training with stimulation may be an option to further enhance its efficacy. Finally, the antisaccadic training might be introduced or modified to be used in the treatment of other eating and weight disorders that are associated with overeating or binge eating, for example, bulimia nervosa.

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